

KOTZEBUE, AK, USA

WMO: 701330

Lat: 66.886N

Lon: 162.613W

Elev: 30

StdP: 14.68

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26616

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -33.1	(c) -28.7	(d) -34.3	(e) 0.8	(f) -27.6	(g) -30.8	(h) 1.0	(i) -24.3	(j) 39.6	(k) 13.6	(l) 36.0	(m) 12.4	(n) 4.9	(o) 60	(p) 0.931

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.6	68.9	60.1	65.4	58.0	62.7	56.7	61.2	67.2	59.2	64.5	57.3	61.9	9.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
58.4	73.0	64.4	56.4	67.7	62.3	54.5	63.2	60.4	27.2	67.1	25.9	64.4	24.7	62.0	70.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 34.2	(b) 29.3	(c) 26.2	(e) -35.9	(f) 74.6	(g) 6.3	(h) 4.3	(i) -40.4	(j) 77.7	(k) -44.1	(l) 80.2	(m) -47.7	(n) 82.6	(o) -52.2	(p) 85.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	24.5	-1.5	2.5	2.1	16.6	33.3	47.5	55.3	52.4	43.7	27.0	11.0	2.7
	DBStd	23.56	17.32	15.90	13.44	11.89	8.97	8.23	6.03	5.68	6.50	9.54	12.22	13.64
	HDD50	9689	1597	1329	1484	1003	521	142	19	38	205	714	1169	1468
	HDD65	14788	2062	1749	1949	1453	983	526	304	391	640	1179	1619	1933
	CDD50	384	0	0	0	0	3	68	185	113	15	0	0	0
	CDD65	8	0	0	0	0	0	2	5	1	0	0	0	0
	CDH74	13	0	0	0	0	0	6	7	0	0	0	0	0
	CDH80	1	0	0	0	0	0	1	0	0	0	0	0	0
Wind	WSAvg	11.4	11.5	12.6	10.4	10.5	9.7	10.5	11.0	11.6	11.7	12.6	12.8	11.8
Precipitation	PrecAvg	9.8	0.6	0.5	0.4	0.4	0.4	0.6	1.5	2.0	1.5	0.9	0.6	0.6
	PrecMax	17.4	1.8	1.2	2.2	1.9	1.4	1.8	3.5	5.3	4.3	2.4	2.2	1.9
	PrecMin	5.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.1
	PrecStd	2.6	0.4	0.3	0.4	0.4	0.3	0.4	0.8	1.3	0.9	0.6	0.5	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.3	34.5	35.3	40.7	60.0	72.8	74.8	71.1	60.9	46.9	36.2	34.0
		MCWB	33.3	32.8	33.3	36.3	49.2	61.1	62.5	61.2	52.1	40.3	34.0	32.5
	2%	DB	31.5	32.0	30.6	37.4	52.7	66.7	70.1	65.9	57.1	43.1	33.9	30.7
		MCWB	29.9	30.6	29.2	33.8	45.4	57.8	60.5	58.6	51.2	39.7	32.5	29.2
	5%	DB	27.2	28.7	26.2	35.3	47.9	63.1	66.4	63.0	54.4	40.8	31.7	26.6
		MCWB	25.8	27.5	24.8	32.9	42.4	56.0	59.1	57.4	49.8	38.4	30.4	25.2
	10%	DB	23.4	24.3	22.1	33.6	43.9	59.3	63.7	60.6	52.2	38.5	28.4	21.9
		MCWB	22.2	23.0	20.9	31.9	39.9	53.1	57.8	55.8	48.6	36.5	27.0	20.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.5	33.3	33.4	36.8	49.7	62.7	64.4	63.1	54.0	42.7	34.3	32.8
		MCDB	35.0	34.2	35.0	40.2	58.3	70.1	71.8	68.2	58.3	44.4	35.5	33.7
	2%	WB	30.2	30.7	29.4	34.4	46.3	58.8	61.9	60.1	52.1	40.5	32.7	29.4
		MCDB	31.6	31.7	30.6	36.5	52.3	65.3	68.2	64.8	55.6	42.3	33.7	30.6
	5%	WB	26.3	27.7	24.9	33.2	42.7	56.2	60.0	58.2	50.6	38.7	30.5	25.4
		MCDB	27.5	28.9	26.2	35.0	47.3	62.2	65.6	62.2	53.7	40.6	31.6	26.8
	10%	WB	22.6	23.3	20.9	31.9	40.0	53.6	58.2	56.3	49.2	36.4	27.2	21.0
		MCDB	23.9	24.4	22.1	33.8	43.9	59.2	63.2	60.1	52.0	38.4	28.5	22.0
Mean Daily Temperature Range	5% DB	MDBR	11.6	12.7	13.4	13.5	10.4	10.2	8.6	7.9	8.0	7.1	8.8	10.9
		MCDBR	12.3	13.7	13.0	11.4	15.8	14.2	12.5	11.4	10.1	6.9	7.5	10.3
		MCWBR	11.8	13.4	12.3	9.6	10.1	9.0	7.6	6.8	6.2	5.4	7.3	9.9
	5% WB	MCDBR	12.6	13.4	13.1	10.9	15.0	12.7	11.1	9.9	8.4	6.2	7.3	10.9
		MCWBR	12.0	13.2	12.6	9.4	9.9	8.7	7.5	6.4	5.6	5.5	7.3	10.5
Clear-Sky Solar Irradiance	taub	0.172	0.238	0.257	0.319	0.357	0.338	0.350	0.360	0.304	0.254	N/A	N/A	
	taud	1.899	2.111	2.224	2.083	2.106	2.399	2.465	2.459	2.547	2.377	N/A	N/A	
	Ebn at Noon	67	193	257	262	263	275	268	252	244	194	N/A	0	
	Edh at Noon	4	16	24	37	41	32	29	26	19	12	N/A	0	
All-Sky Solar Radiation	RadAvg	21	191	750	1408	1823	1974	1529	1115	690	246	44	0	
	RadStd	3	26	65	95	170	141	139	139	72	27	7	0	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+2.57	N/A	N/A	+1.80	+1.48	N/A	-855	-918	+117	+2
(47) Regional (8 neighbors)		+2.57	N/A	N/A	+1.80	+1.48	N/A	-855	-918	+117	+2

Nomenclature: See separate page